بسم الله الرحمن الرحيم

«وما أوتيتم من العلم الا قليلا»



Respiratory Failure

By

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The respiratory system supplies the body with adequate oxygen for aerobic metabolism and simultaneously removes its major metabolic waste product carbon dioxide.

Definition of Respiratory Failure

- **Respiratory failure is a syndrome in which the respiratory system fails in one or both of its gas exchange functions: oxygenation and carbon dioxide elimination, due to dysfunction of one or more of the essential components of the respiratory system**

Respiratory System

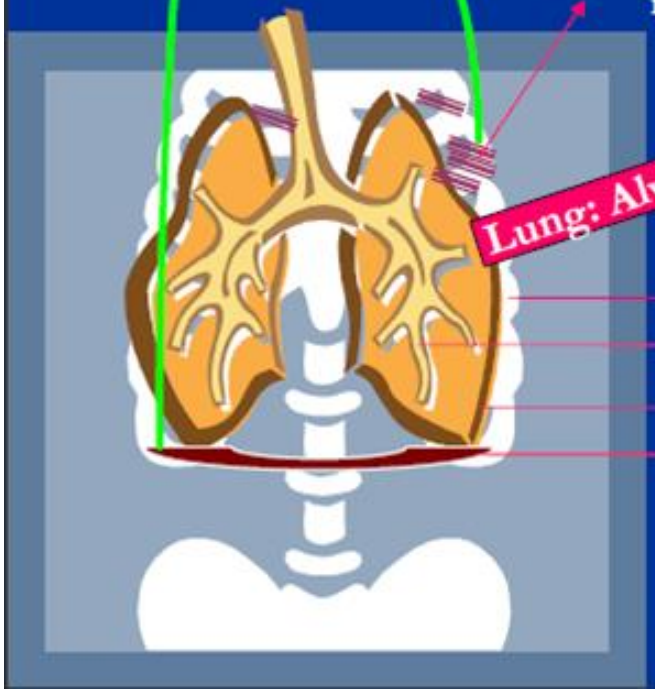


Brain

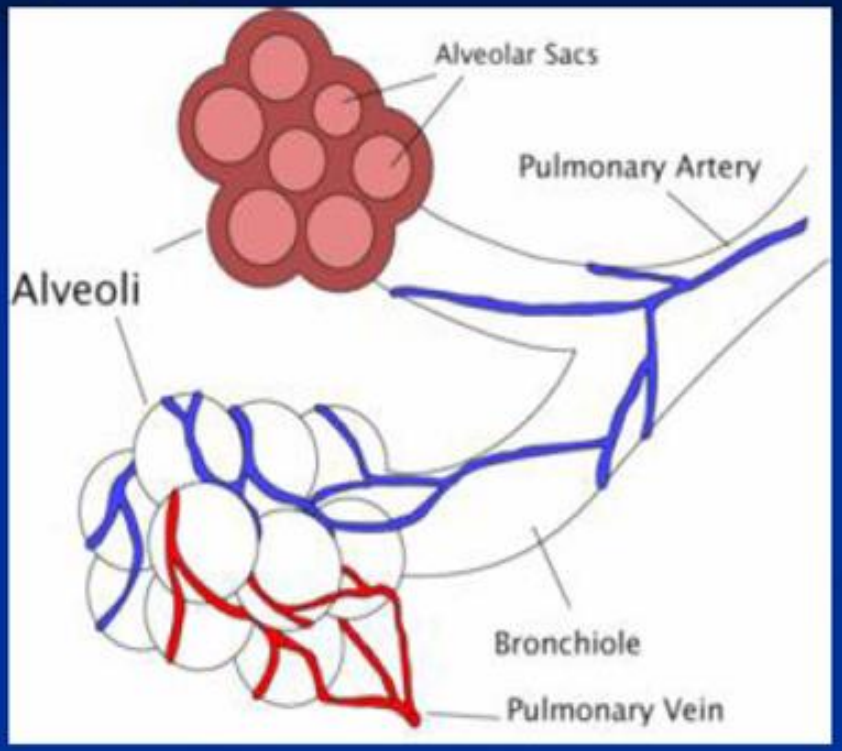
Spinal cord

Nerves

Intercostal muscles



Lung: Alveolar Unit



Chest wall

Airway

Pleura

Diaphragm

- **PO_2 < 60 mmHg**

➢ **At sea Level**

➢ **At rest**

➢ **On room air**

➢ **Rule out other causes (Congenital cyanotic heart disease and abnormal types of hemoglobin)**

Types of respiratory failure

- **Type I respiratory failure (Hypoxemic type):** is characterized by an arterial oxygen tension (PaO_2) lower than 60 mm Hg with a normal or low arterial carbon dioxide tension ($PaCO_2$). This is the most common form of respiratory failure.
- **Type II respiratory failure (Hypercapnic type):** characterized by PaO_2 lower than 60 mm Hg with increased $PaCO_2$.

- **Type 1:**

  $PaO_2 < 60$ mm Hg

  $PaCo_2 \leq 35\text{-}45$ mmHg

- **Type 2:**

  $PaO_2 < 60$ mm Hg

  $PaCo_2 > 50$ mmHg

Handerson Hasselbach Equation

- **PH= pKa+HCO3/PaCo2**
- **PH= HCO3/PaCo2**
- **↑PH= ↑ HCO3**
- **↓PH= ↓ HCO3**
- **↑ PH= ↓ PaCo2**
- **↓PH= ↑ PaCo2**

What is PH?

- PH is the negative log of the activity of the hydrogen ion in an aqueous solution.

Type 1:

- **Acute:**

Pao2<60mmHg HCO3 Normal
PaCo2⊆35 mmHg PH > 7.45

- **Chronic:**

Pao2<60mmHg PaCo2⊆35mmHg
HCO3 <22mEq/L PH=7.40-7.45

Type 2:

- **Acute:**

Pao2<60mmHg HCO3 Normal

PaCo2>50 mmHg PH<7.35

- **Chronic:**

Pao2<60mmHg PaCo2 >50 mmHg

HCO3 >26 mEq/L PH=7.35-7.40

Type 1 RF

	PH	pO2	pCO2	HCO3
Acute	↑	↓	↓	**normal**
Chronic	**normal**	↓	↓	↓

Type 2 RF

	PH	pO2	pCO2	HCO3
Acute	↓	↓	↑	**normal**
Chronic	**normal**	↓	↑	↑

Aetiology
ACUTE TYPE 1 RF

- **Cardiogenic pulmonary edema**
- **Non cardiogenic pulmonary edema**
- **Pneumonia**
- **Pulmonary Embolism**
- **Interstitial Lung Disease**
- **Severe bronchial asthma without diaphragmatic fatigue**
- **Acute milliary TB and lymphagitis tuberculosa reticularis**

ChronicType I RF

- **Fibrosing alveolitis**
- **Other causes of IPF**
- **Chronic allergic alveolitis**
- **Chronic thromboembolic pulmonary hypertension**
- **Chronic pulmonary edema**
- **Lymphangitis carcinomatosis**

Acute Type II RF

- **Upper airway obstruction**
- **Acute severe asthma with diaphragmatic fatigue**
- **Acute CNS disorder**
- **Myathenia gravis**
- **polyneuritis**
- **Acute Exacerbation of COPD**

Chronic Type II RF

- **COPD**
- **Obesity Hypoventilation Syndrome**
- **kyphoscoliosis**
- **Chronic neuromuscular diseases**
- **Progressive respiratory diseases preterminal**

Diagnosis

- **Clinical picture of causative disease**
- **Manifestations of hypoxaemia.**

1-Central cyanosis if reduced hemoglobin is >5gram%.

2- Restlessness, irritability,, impaired intellectual functions.

3- Acute severe hypoxaemia may cause convulsions, coma and death.

4- Hyperventilation and tachypnae through stimulation of chemoreceptors

5- Tachycardia, arrythmias, increased COP and dilatation of peripheral vessels

6- Pulmonary vasoconstriction with pulmonary hypertension

7- Secondary polycythaemia with predisposition to DVT and pulmonary embolism

- **Manifestations of hypercapnia:**

1-Drowsines, flapping tremors, coma (CO_2 narcosis) and papillodema due to increased CSF formation secondary to cerebral vasodilatation and increased cerebral blood flow.

2-Paradoxical action on peripheral blood vessels: Vasodilatation through direct action and vasoconstriction through sympathetic stimulation and the predominant action is the local one.

3-Tachycardia, sweating and generalized vasodilatation with hypotension.

Investigations

- **Investigations for the underlying cause**
- **Arterial blood gas analysis (ABG).**
- **Non-invasive methods**

Management

- **Treatment of the underlying cause**
- **Correction of hypoxaemia**
- **Treatment of complications**

Complications of RF:

1-Cardiac arrythmias due to severe hypoxaemia and acidaemia secondary to CO2 retention.

2-Pulmonary hypertension and cor pulmonale due to pulmonary vasoconstriction as a result of hypoxaemia and acidaemia.

3-DVT and pulmonary embolism due to polycythaemia secondary to chronic hypoxaemia.

4-Complications of oxygen therapy and mechanical ventilation.

Criteria for acute type 1 respiratory failure include the following except:

1-PH 7.54

2-PaO_2 44 mmHg

3-$PaCO_2$ 24 mmHg

4-HCO_3 36 mEq/L

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Criteria for chronic type 1 respiratory failure include the following except:

1-PH 7.30

2-PaO_2 50 mmHg

3-$PaCO_2$ 30 mmHg

4-HCO_3 16 mEq/L

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1-PH 7.30

2-PaO2 50 mmHg

3-PaCO2 30 mmHg

4-HCO3 16 mEq/L

Criteria for chronic type 11 respiratory failure include the following except:

1-PH 7.38

2-PaO2 48 mmHg

3-PaCO2 70mmHg

4-HCO3 15 mEq/L

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1-PH 7.38

2-PaO_2 48 mmHg

3-$PaCO_2$ 70mmHg

4-HCO_3 15 mEq/L

Criteria for acute type 11 respiratory failure include the following except:

1-PH 7.60

2-PaO_2 40mmHg

3-$PaCO_2$ 80mmHg

4-HCO_3 25 mEq/L

Criteria for acute type 11 respiratory failure include the following except:

1-PH 7.60

2-PaO2 40mmHg

3-PaCO2 80mmHg

4-HCO3 25 mEq/L

Clinical evidence of hypoxaemia include the following except:

1-Central cyanosis
2-Cardiac arrhythmias
3-Pulmonary hypertension
4-Tachypnae
5-Gastric distension and decreased intestinal motility

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1-Disturbed conscious level
2-Central cyanosis
3- Gastric distension and decreased intestinal motility
4-Flapping tremors
5-Papilloedema

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Thank
You